QUESTION BANK

By

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SOFT COMPUTING

B.TECH. CSE VIII SEM (ELECTIVE)

UNIT-1

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1. Define Soft Computing and Compare it with Hard Computing.

Soft computing, as opposed to traditional computing, deals with approximate models and gives solutions to complex real-life problems. Soft computing is tolerant of imprecision, uncertainty, partial truth, and approximations. In effect, the role model for soft computing is the human mind.

The principal constituents of Soft Computing (SC) are Fuzzy Logic (FL), Neural Computing (NC), Evolutionary Computation (EC) Machine Learning (ML) and Probabilistic Reasoning (PR), with the latter subsuming belief networks, chaos theory and parts of learning theory.

Soft computing vs. hard computing:

Following points clearly differentiate the both:

1. Soft Computing is tolerant of imprecision, uncertainty, partial truth and approximation whereas Hard Computing requires a precisely state analytic model.

2. Soft Computing is based on fuzzy logic, neural sets, and probabilistic reasoning whereas hard Computing is based on binary logic, crisp system, numerical analysis and crisp software.

3. Soft computing has the characteristics of approximation and dispositional whereas hard computing has the characteristics of precision and categorical.

4. Soft computing can evolve its own programs whereas hard computing requires programs to be written.

5. Soft computing can use multi valued or fuzzy logic whereas hard computing uses two-valued logic.

2. Explain various types of Soft Computing Techniques.

Soft computing differs from hard (conventional) computing. Unlike hard computing, the soft computing is tolerant of imprecision, uncertainty, partial truth, and approximation. The guiding

principle of soft computing is to exploit these tolerance to achieve tractability, robustness and low solution cost. In effect, the role model for soft computing is the human mind.

The four fields that constitute Soft Computing (SC) are: Fuzzy Computing (FC),Evolutionary Computing (EC), Neural computing (NC), and Probabilistic Computing (PC), with the latter subsuming belief networks, chaos theory and parts of learning theory.

Soft computing is not a concoction, mixture, or combination; rather, Soft computing is a partnership in which each of the partners contributes a distinct methodology for addressing problems in its domain. In principal the constituent methodologies in Soft computing are complementary rather than competitive.

Soft computing may be viewed as a foundation component for the emerging field of Conceptual Intelligence

Various types of soft computing techniques:

Soft computing is a fusion of three different methodologies as follows:

Soft computing = Fuzzy Logic + Neural Network + Genetic Algorithm

Brief details about above mentioned techniques are as follows:

1. Fuzzy Logic:

Fuzzy logic is intended to model logical reasoning with vague or imprecise statements like “Petr is young (rich, tall, hungry, etc.)”. It refers to a family of many-valued logics (see entry on many-valued logic) and thus stipulates that the truth value (which, in this case amounts to a degree of truth) of a logically compound proposition, like “Carles is tall and Chris is rich”, is determined by the truth value of its components. In other words, like in classical logic, one imposes truth-functionality.

Fuzzy logic emerged in the context of the theory of fuzzy sets, introduced by Zadeh (1965). A fuzzy set assigns a degree of membership, typically a real number from the interval [0,1] , to elements of a universe. Fuzzy logic arises by assigning degrees of truth to propositions. The standard set of truth values (degrees) is [0,1], where 0 represents “totally false”, 1 represents “totally true”, and the other numbers refer to partial truth, i.e., intermediate degrees of truth.

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"Fuzzy logic" is often understood in a very wide sense which includes all kinds of formalisms and techniques referring to the systematic handling of degrees of some kind. In particular in engineering contexts (fuzzy control, fuzzy classification, soft computing) it is aimed at efficient computational methods tolerant to sub optimality and imprecision. It focuses on logics based on a truth-functional account of partial truth and studies them in the spirit of classical mathematical logic (syntax, model theoretic semantics, proof systems, completeness, etc.; both, at propositional and the predicate level).

2. Neural Network:

Neural Networks (NNs) are also known as Artificial Neural Networks (ANNs), Connectionist Models, and Parallel Distributed Processing (PDP) Models. Artificial Neural Networks are massively parallel interconnected networks of simple (usually adaptive) elements and their hierarchical organizations which are intended to interact with the objects of the real world in the same way as biological nervous systems do. Fine-grained, parallel, distributed computing model characterized by:

- A large number of very simple, neuron-like processing elements called units, PEs, or nodes
- A large number of weighted, directed connections between pairs of units
- Weights may be positive or negative real values
- Local processing in that each unit computes a function based on the outputs of a limited number of other units in the network
- Each unit computes a simple function of its input values, which are the weighted outputs from other units. If there are n inputs to a unit, then the unit's output, or activation is defined by $a = g((w1 * x1) + (w2 * x2) + ... + (wn * xn))$. Thus each unit computes a (simple) function g of the linear combination of its inputs.

3. Genetic Algorithm:

Genetic algorithm is basically a randomized search by simulating evolution, starting from an initial set of solutions or hypotheses, and generating successive "generations" of solutions. This particular branch of AI was inspired by the way living things evolved into more successful organisms in nature. The main idea is survival of the fittest, a.k.a. natural selection.

A chromosome is a long, complicated thread of DNA (deoxyribonucleic acid). Hereditary factors that determine particular traits of an individual are strung along the length of these chromosomes, like beads on a necklace. Each trait is coded by some combination of DNA (there are four bases, A (Adenine), C (Cytosine), T (Thymine) and G (Guanine). Like an alphabet in a language, meaningful combinations of the bases produce specific instructions to the cell.

Changes occur during reproduction. The chromosomes from the parents exchange randomly by a process called crossover. Therefore, the offspring exhibit some traits of the father and some traits of the mother.

A rarer process called mutation also changes some traits. Sometimes an error may occur during copying of chromosomes (mitosis). The parent cell may have -A-C-G-C-T- but an accident may occur and changes the new cell to -A-C-T-C-T-. Much like a typist copying a book, sometimes a few mistakes are made. Usually this results in a nonsensical word and the cell does not survive.

3. Define Fuzzy logic with example. Write comparison between fuzzy logic and crisp logic.

Fuzzy logic is intended to model logical reasoning with vague or imprecise statements like "Petr is young (rich, tall, hungry, etc.)". It refers to a family of many-valued logics (see entry on many-valued logic) and thus stipulates that the truth value (which, in this case amounts to a degree of truth) of a logically compound proposition, like "Carles is tall and Chris is rich", is determined by the truth value of its components. In other words, like in classical logic, one imposes truth-functionality.

Fuzzy logic emerged in the context of the theory of fuzzy sets, introduced by Zadeh (1965). A fuzzy set assigns a degree of membership, typically a real number from the interval [0,1] , to elements of a universe. Fuzzy logic arises by assigning degrees of truth to propositions. The standard set of truth values (degrees) is [0,1], where 0 represents "totally false", 1 represents "totally true", and the other numbers refer to partial truth, i.e., intermediate degrees of truth.

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Crisp and Non-crisp Set

– As said before, in classical set theory, the **characteristic function** $\propto_A(x)$ of Eq.(2) has only values **0** ('false') and **1** ('true'').

Such sets are **crisp sets.**

– For Non-crisp sets the characteristic function $\propto_A(x)$ can be defined.

- The characteristic function $\propto_A(x)$ of Eq. (2) for the crisp set is generalized for the Non-crisp sets.
- This generalized characteristic function $\propto_A(x)$ of Eq.(2) is called **membership function**.

Such Non-crisp sets are called **Fuzzy Sets**.

– Crisp set theory is not capable of representing descriptions and classifications in many cases; In fact, Crisp set does not provide adequate representation for most cases.

– The proposition of Fuzzy Sets are motivated by the need to capture and represent real world data with uncertainty due to imprecise measurement.

– The uncertainties are also caused by vagueness in the language.

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Example 1 : Heap Paradox

This example represents a situation where vagueness and uncertainty are inevitable.

- If we remove one grain from a heap of grains, we will still have a heap.
- However, if we keep removing one-by-one grain from a heap of grains, there will be a time when we do not have a heap anymore.
- The question is, at what time does the heap turn into a countable collection of grains that do not form a heap? There is no one correct answer to this question.

Example 2 : Classify Students for a basketball team This example explains the grade of truth value.

- **tall students** qualify and **not tall students** do not qualify
- if students 1.8 m tall are to be qualified, then

  should we exclude a student who is $^{1}/10$" less? or should we exclude a student who is 1" shorter?

- Non-Crisp Representation to represent the notion of a tall person.

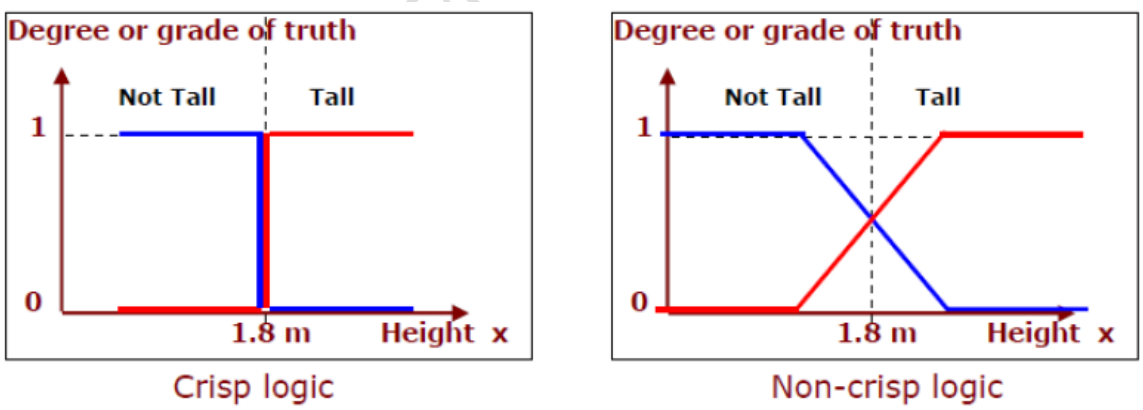

Fig. 1 Set Representation – Degree or grade of truth

- A fuzzy set is described by a **membership function** $\mu_A(x)$ of **A**.
  This membership function associates to each element $x_\sigma \in X$ a number as $\mu_A(x_\sigma)$ in the closed unit interval **[0, 1]**.

  The number $\mu_A(x_\sigma)$ represents the **degree of membership** of x_σ in **A**.
- The notation used for membership function $\mu_A(x)$ of a fuzzy set **A** is

  $A : X \rightarrow [0, 1]$
- Each membership function maps elements of a given universal base set **X** , which is itself a crisp set, into real numbers in **[0, 1]** .
- Example

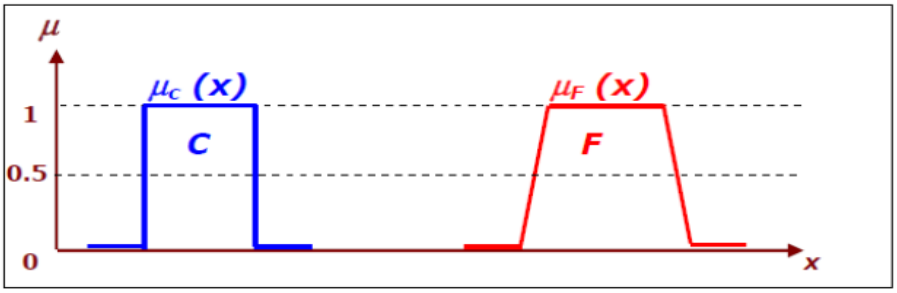

Fig. 2 Membership function of a Crisp set C and Fuzzy set F

- In the case of Crisp Sets the members of a set are :
  either out of the set, with membership of degree " **0** ", or in the set, with membership of degree " **1** ",

Therefore, **Crisp Sets** ⊆ **Fuzzy Sets** In other words, Crisp Sets are Special cases of Fuzzy Sets.

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Example 2: Set of SMALL (as non-crisp set) **Example 1: Set of prime numbers** (a crisp set)

If we consider space **X** consisting of natural numbers ≤ **12**

ie **X = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12}**

Then, the set of prime numbers could be described as follows.

PRIME = {x contained in X | x is a prime number} = {2, 3, 5, 6, 7, 11}

A Set **X** that consists of SMALL cannot be described;

for example **1** is a member of SMALL and **12** is not a member of SMALL.

Set **A**, as SMALL, has un-sharp boundaries, can be characterized by a function that assigns a real number from the closed interval from **0** to **1** to each element **x** in the set **X**.

4. **Compare classical and fuzzy sets.**

Classical Set Theory

- **Sets** are defined by a simple statement describing whether an element having a certain property belongs to a particular set.
- When set **A** is contained in an universal space **X**,

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then we can state explicitly whether each element **x** of space **X** "is or is not" an element of **A**.

Set **A** is well described by a function called **characteristic function A**. This function, defined on the universal space **X**, assumes :

value **1** for those elements **x** that belong to set **A**, and

value **0** for those elements **x** that do not belong to set **A**.

The notations used to express these mathematically are

$$A : X \rightarrow [0, 1]$$

$$\left.\begin{array}{l} A(x) = 1 \text{ , } x \text{ is a member of } A \\ A(x) = 0 \text{ , } x \text{ is not a member of } A \end{array}\right\} \quad \text{Eq.(1)}$$

Alternatively, the set **A** can be represented for all elements $x \in X$ by its **characteristic function** $\propto_A (x)$ defined as

$$\propto_A (x) = \begin{cases} 1 & \text{if } x \in X \\ 0 & \text{otherwise} \end{cases} \quad \text{Eq.(2)}$$

Thus, in classical set theory $\propto_A (x)$ has only the values **0** ('false') and **1** ('true''). Such sets are called **crisp sets.**

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Definition of Fuzzy Set

A **fuzzy set A** defined in the universal space **X** is a function defined in **X** which assumes values in the range **[0, 1]**.

A fuzzy set **A** is written as a set of pairs **{x, A(x)}** as

A = {{x , A(x)}} , x in the set X

where **x** is an element of the universal space **X**, and
A(x) is the value of the function **A** for this element.

The value **A(x)** is the **membership grade** of the element **x** in a fuzzy set **A**.

Example : Set **SMALL** in set **X** consisting of natural numbers **to 12**.

Assume:

SMALL(1) = 1, SMALL(2) = 1, SMALL(3) = 0.9, SMALL(4) = 0.6,

SMALL(5) = 0.4, SMALL(6) = 0.3, SMALL(7) = 0.2, SMALL(8) = 0.1,
SMALL(u) = 0 for u >= 9.

Then, following the notations described in the definition above :

Set SMALL = {{1, 1 }, {2, 1 }, {3, 0.9}, {4, 0.6}, {5, 0.4}, {6, 0.3}, {7, 0.2},
{8, 0.1}, {9, 0 }, {10, 0 }, {11, 0}, {12, 0}}

Note that a fuzzy set can be defined precisely by associating with each **x** , its grade of membership in **SMALL**.

5. **Explain Various Fuzzy Operations.**

Fuzzy Operations

A fuzzy set operations are the operations on fuzzy sets. The fuzzy set operations are generalization of crisp set operations. Zadeh [1965] formulated the fuzzy set theory in the terms of standard operations: Complement, Union, Intersection, and Difference.

In this section, the graphical interpretation of the following standard fuzzy set terms and the Fuzzy Logic operations are illustrated:

Inclusion : **FuzzyInclude [VERYSMALL, SMALL]**

Equality : **FuzzyEQUALITY [SMALL, STILLSMALL]**

Complement : **FuzzyNOTSMALL = FuzzyCompliment [Small]**

Union : **FuzzyUNION = [SMALL ∪ MEDIUM]**

Intersection : **FUZZYINTERSECTON = [SMALL ∩ MEDIUM]**

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- **Inclusion**

Let **A** and **B** be fuzzy sets defined in the same universal space **X**.

The fuzzy set **A** is included in the fuzzy set **B** if and only if for every **x** in

the set **X** we have **A(x)** ≤ **B(x)**

Example :

The fuzzy set **UNIVERSALSPACE** numbers, defined in the universal space **X = { x_i } =** {1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12} is presented as

SetOption [FuzzySet, UniversalSpace → {1, 12, 1}]

The fuzzy set B SMALL

The Set **SMALL** in set **X** is :

SMALL = FuzzySet {{1, 1 }, {2, 1 }, {3, 0.9}, {4, 0.6}, {5, 0.4}, {6, 0.3}, {7, 0.2}, {8, 0.1}, {9, 0 }, {10, 0 }, {11, 0}, {12, 0}}

Therefore **SetSmall** is represented as

SetSmall = FuzzySet [{{1,1},{2,1}, {3,0.9}, {4,0.6}, {5,0.4},{6,0.3}, {7,0.2}, {8, 0.1}, {9, 0}, {10, 0}, {11, 0}, {12, 0}} , **UniversalSpace →** {1, 12, 1}]

The fuzzy set A VERYSMALL

The Set **VERYSMALL** in set **X** is :

VERYSMALL = FuzzySet {{1, 1 }, {2, 0.8 }, {3, 0.7}, {4, 0.4}, {5, 0.2}, {6, 0.1}, {7, 0 }, {8, 0 }, {9, 0 }, {10, 0 }, {11, 0}, {12, 0}}

Therefore **SetVerySmall** is represented as

SetVerySmall = FuzzySet [{{1,1},{2,0.8}, {3,0.7}, {4,0.4}, {5,0.2},{6,0.1}, {7,0}, {8, 0}, {9, 0}, {10, 0}, {11, 0}, {12, 0}} , **UniversalSpace →** {1, 12, 1}]

The Fuzzy Operation :

nclusion

Include

[VERYSMALL, SMALL]

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- **Comparability**

Two fuzzy sets **A** and **B** are comparable

if the condition **A** $\subset$ **B or B** $\subset$ **A** holds, ie,

if one of the fuzzy sets is a subset of the other set, they are comparable.

Two fuzzy sets **A** and **B** are incomparable

if the condition **A** $\not\subset$ **B or B** $\not\subset$ **A** holds.

Example 1:

Let **A = {{a, 1}, {b, 1}, {c, 0}}** **and**

B = {{a, 1}, {b, 1}, {c, 1}}.

Then **A** is comparable to **B**, since **A** is a subset of **B**.

Example 2 :

Let **C = {{a, 1}, {b, 1}, {c, 0.5}}** **and**

D = {{a, 1}, {b, 0.9}, {c, 0.6}}.

Then **C** and **D** are not comparable since

C is not a subset of **D** and

D is not a subset of **C**.

Property Related to Inclusion :

for all **x** in the set **X**, if **A(x)** $\subset$ **B(x)** $\subset$ **C(x), then accordingly A** $\subset$ **C.**

- **Equality**

Let **A** and **B** be fuzzy sets defined in the same space **X**.
Then **A** and **B** if
and only if are equal, which is denoted **X = Y**

for all **x** in the set **X**, **A(x) = B(x).**

Example.

The fuzzy set B SMALL

SMALL = FuzzySet {{1, 1 }, {2, 1 }, {3, 0.9}, {4, 0.6}, {5, 0.4}, {6, 0.3},
{7, 0.2}, {8, 0.1}, {9, 0 }, {10, 0 }, {11, 0}, {12, 0}}

The fuzzy set A STILLSMALL

STILLSMALL = FuzzySet {{1, 1 }, {2, 1 }, {3, 0.9}, {4, 0.6}, {5, 0.4},

{6, 0.3}, {7, 0.2}, {8, 0.1}, {9, 0 }, {10, 0 }, {11, 0}, {12, 0}}

The Fuzzy Operation : Equality

Equality [SMALL, STILLSMALL]

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the set **X**, then we say that **A** is not equal to **B**.

- **Complement**

Let **A** be a fuzzy set defined in the space **X**.

Then the fuzzy set **B** is a complement of the fuzzy set **A**, if and only if,

for all **x** in the set **X**, **B(x) = 1 - A(x).**

The complement of the fuzzy set **A** is often denoted by **A'** or A_c or $\overline{A}$

Fuzzy Complement : $A_c(x) = 1 - A(x)$

Example 1.

The fuzzy set A SMALL

SMALL = FuzzySet {{1, 1 }, {2, 1 }, {3, 0.9}, {4, 0.6}, {5, 0.4}, {6, 0.3}, {7, 0.2}, {8, 0.1}, {9, 0 }, {10, 0 }, {11, 0}, {12, 0}}

The fuzzy set Ac NOTSMALL

NOTSMALL = FuzzySet {{1, 0 }, {2, 0 }, {3, 0.1}, {4, 0.4}, {5, 0.6}, {6, 0.7}, {7, 0.8}, {8, 0.9}, {9, 1 }, {10, 1 }, {11, 1}, {12, 1}}

The Fuzzy Operation : Compliment

NOTSMALL = Compliment [SMALL]

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Example 2.

The empty set Φ and the universal set **X**, as fuzzy sets, are complements of one another.

$$\Phi' = X \quad , \quad X' = \Phi$$

The fuzzy set B EMPTY

Empty = FuzzySet {{1, 0 }, {2, 0 }, {3, 0}, {4, 0}, {5, 0}, {6, 0},

{7, 0}, {8, 0}, {9, 0 }, {10, 0 }, {11, 0}, {12, 0}}

The fuzzy set A UNIVERSAL

Universal = FuzzySet {{1, 1 }, {2, 1 }, {3, 1}, {4, 1}, {5, 1}, {6, 1},

{7, 1}, {8, 1}, {9, 1 }, {10, 1 }, {11, 1}, {12, 1}}

The fuzzy operation : Compliment

EMPTY = Compliment [UNIVERSALSPACE]

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- **Union**

Let **A** and **B** be fuzzy sets defined in the space **X**.

The union is defined as the smallest fuzzy set that contains both **A** and **B**. The union of **A** and **B** is denoted by **A** ∪ **B.**

The following relation must be satisfied for the union operation
: **for all x in the set X, (A ∪ B)(x) = Max (A(x), B(x)).**

Fuzzy Union : **(A ∪ B)(x) = max [A(x), B(x)] for all x ∈ X**

Example 1 : Union of Fuzzy **A** and **B**

A(x) = 0.6 and B(x) = 0.4 ∴ (A ∪ B)(x) = max [0.6, 0.4] = 0.6

Example 2 : Union of **SMALL** and **MEDIUM**

The fuzzy set A SMALL

SMALL = FuzzySet {{1, 1 }, {2, 1 }, {3, 0.9}, {4, 0.6}, {5, 0.4}, {6, 0.3}, {7, 0.2}, {8, 0.1}, {9, 0 }, {10, 0 }, {11, 0}, {12, 0}}

The fuzzy set B MEDIUM

MEDIUM = FuzzySet {{1, 0 }, {2, 0 }, {3, 0}, {4, 0.2}, {5, 0.5}, {6, 0.8}, {7, 1}, {8, 1}, {9, 0.7 }, {10, 0.4 }, {11, 0.1}, {12, 0}}

The fuzzy operation : Union

FUZZYUNION = [SMALL ∪ MEDIUM]

SetSmallUNIONMedium = FuzzySet [{{1,1},{2,1}, {3,0.9}, {4,0.6}, {5,0.5}, {6,0.8}, {7,1}, {8, 1}, {9, 0.7}, {10, 0.4}, {11, 0.1}, {12, 0}} , **UniversalSpace →** {1, 12, 1}]

The notion of the union is closely related to that of the connective "or".

Let **A** is a class of "Young" men, **B** is a class of "Bald" men.

- **Intersection**

Let **A** and **B** be fuzzy sets defined in the space **X**.

The intersection is defined as the greatest fuzzy set included both **A** and **B**. The intersection of **A** and **B** is denoted by **A** ∩ **B.**

The following relation must be satisfied for the union operation :

for all x in the set X, (A ∩ B)(x) = Min (A(x), B(x)).

Fuzzy Intersection : (A ∩ B)(x) = min [A(x), B(x)] **for all x ∈ X**

Example 1 : Intersection of Fuzzy **A** and **B**

A(x) = 0.6 and B(x) = 0.4 ∴ (A ∩ B)(x) = min [0.6, 0.4] = 0.4

Example 2 : Union of **SMALL** and **MEDIUM**

The fuzzy set A SMALL

SMALL = FuzzySet {{1, 1 }, {2, 1 }, {3, 0.9}, {4, 0.6}, {5, 0.4}, {6, 0.3},

{7, 0.2}, {8, 0.1}, {9, 0 }, {10, 0 }, {11, 0}, {12, 0}}

The fuzzy set B MEDIUM

MEDIUM = FuzzySet {{1, 0 }, {2, 0 }, {3, 0}, {4, 0.2}, {5, 0.5}, {6, 0.8},

{7, 1}, {8, 1}, {9, 0.7 }, {10, 0.4 }, {11, 0.1}, {12, 0}}

The fuzzy operation : Intersection FUZZYINTERSECTION = min [SMALL ∩ MEDIUM] SetSmallINTERSECTIONMedium = FuzzySet [{{1,0},{2,0}, {3,0}, {4,0.2},

{5,0.4}, {6,0.3}, {7,0.2}, {8, 0.1}, {9, 0},{10, 0}, {11, 0}, {12, 0}} , **UniversalSpace →** {1, 12, 1}]

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